Findings of the Study

The study gathered survey responses from 42 elderly residents and 69 caregivers at Bleu Heritage Nursing Home and Loving Home Nursing Care. Of the elderly respondents, 19 were from Bleu Heritage and 23 were from Loving Home Nursing Care. The caregiver group comprised 28 respondents from Bleu Heritage and 41 from Loving Home Nursing Care. These findings describe the participating groups and provide context for the proposed ElderCareAI system rather than representing all residential care facilities.

The results identified both established monitoring practices and difficulties in the existing care process. Elderly respondents strongly agreed that caregivers regularly checked their health vitals (M = 4.33), while their response to delays in urgent medical attention was neutral (M = 2.95). Caregivers strongly agreed that they felt overwhelmed by simultaneously monitoring several patients (M = 4.61) and that manual monitoring was time-consuming (M = 4.52). Their ability to monitor every patient's vitals received a neutral rating (M = 3.04). However, they also strongly agreed that existing documentation and reporting were efficient (M = 4.51). These responses indicate specific monitoring burdens without establishing that every part of the existing process is inadequate.

Technology familiarity and wearable comfort supported the relevance of an accessible mobile interface. Among caregivers, 55.1% described themselves as familiar and 23.2% as very familiar with work-related mobile applications. Among elderly respondents, 45.2% were comfortable and 33.3% were very comfortable wearing a smartwatch. Nevertheless, respondents also agreed with concern about wearing a monitoring device throughout the day (M = 4.14). Thus, favorable attitudes toward wearable monitoring coexist with practical concerns that should inform training, fit and daily use.

The monitoring-difficulty results highlighted stress level, selected by 53 caregivers (76.8%), followed by sleep pattern, selected by 43 (62.3%), and skin temperature, selected by 34 (49.3%). These multiple-response findings identify areas where caregivers reported a need for support; they do not demonstrate that a wearable can clinically assess those conditions.

Caregivers strongly agreed with the proposed system mechanisms overall (reported grand mean = 4.53), with chatbot guidance receiving the highest item mean of 4.64. The proposed key features received a reported grand mean of 4.58. Four features shared the highest mean of 4.61: stress-level information, skin-temperature deviation, AI-based health-risk prediction and automated health reports. Overall, the survey supports interest in organized monitoring, notifications, AI assistance and reporting, while leaving their actual operational and clinical effects to further evaluation.

CONCLUSION

The study concludes that the participating elderly residents and caregivers identified a practical need for monitoring support and expressed favorable expectations toward ElderCareAI. Regular checks were recognized by elderly respondents, but caregivers reported time demands, difficulty tracking several patients and feelings of being overwhelmed. These findings support developing tools that make available patient information easier to review while preserving effective practices already present in the facilities.

The responses indicate that mobile applications and wearable devices are acceptable to many participants. Caregivers' familiarity with work-related apps and elderly respondents' reported smartwatch comfort provide a basis for introducing a digital monitoring platform. However, the concern about prolonged device use and the presence of respondents unfamiliar with mobile applications show that acceptance should not be assumed for every user. Clear instructions, readable interfaces, assisted onboarding and attention to wrist comfort are necessary considerations for implementation.

ElderCareAI brings Galaxy Watch6 measurements, recorded health history, care schedules, location information and caregiver notifications into a shared application. Elle provides assistance using available care records, while the system offers health summaries and reporting. These capabilities correspond to the monitoring and information needs reflected in the survey. Their use should complement caregivers' observations and existing care protocols, with clear identification of missing readings, delayed uploads and uncertain location estimates.

The evidence supports perceived usefulness and a reason for further implementation, rather than a conclusion that the system has already improved health outcomes, prevented complications or reduced response time. Agreement with proposed benefits is not a measurement of sensor accuracy or clinical effectiveness. Wrist skin temperature is not a fever diagnosis, the HRV-derived stress estimate is experimental, and sleep records depend on synchronization. Predictions and AI guidance should therefore remain supportive information subject to professional judgment.

The findings justify a supervised evaluation of ElderCareAI within the participating care settings. Further assessment should examine actual usability, notification delivery, data completeness, device comfort and integration into daily routines. Broader conclusions about efficiency, safety and long-term benefit require observed use, appropriate comparison and validation beyond the perceptions reported in Chapter IV.

TRANSLATIONAL RESEARCH

The translational direction of ElderCareAI is to turn the survey-identified monitoring needs into a practical, supervised caregiving tool. The intended application is an Android caregiver platform connected to Galaxy Watch6, with recorded health history, reminders, notifications, location information and Elle assistance. The findings provide priorities for implementation: reducing information fragmentation, supporting review of available readings and making important care tasks visible. They do not establish that these benefits have already occurred.

An initial pilot may be proposed at Bleu Heritage Nursing Home and Loving Home Nursing Care, subject to facility approval, participant consent and appropriate data-access arrangements. The team should begin with a limited, manageable set of patients and caregivers. Orientation should cover phone and watch installation, patient-specific pairing, permissions, monitoring controls and notification settings. Staff should also learn how to distinguish the original measurement time from upload time and how to recognize a missing or stale value.

Practical use should follow the available data paths. Supported live measurements come from the watch while monitoring is active, whereas sleep and available overnight oxygen records are imported after synchronization with Samsung Health. Caregivers can manage medication and appointment schedules and review confirmations. Location alerts should identify their source and accuracy because fused estimates can represent the paired phone. The pilot should retain existing clinical monitoring and emergency procedures while staff determine how these functions fit their work.

Evaluation should combine user feedback with observed technical and workflow measures. Suggested measures include the proportion of usable readings, upload delay, notification delivery and acknowledgment, completeness of medication confirmations, time needed to review records, and reported device comfort. Network interruptions, battery limitations, inaccurate locations, false alerts and missed notifications should be documented. These measurements would allow the team to test the proposed benefits instead of treating favorable survey ratings as evidence of performance.

Findings from the pilot can guide improvements to instructions, interface accessibility, alert settings and synchronization reliability. The strong interest in stress, sleep and temperature information should be addressed with explanations of what each measure can and cannot indicate. Larger or longer evaluations should be considered only after technical reliability and user concerns have been addressed. Through this staged process, ElderCareAI can be evaluated as a caregiver-support system whose use remains accountable to patient needs, facility procedures and professional care judgment.

ABSTRACT

ElderCareAI is an AI-assisted Internet of Things system designed to support elderly monitoring and caregiver tasks through an Android application and Galaxy Watch6 companion. This study examined the care context and respondents' perceptions of the proposed system using survey findings from 42 elderly residents and 69 caregivers at Bleu Heritage Nursing Home and Loving Home Nursing Care. Frequencies, percentages and item means were used to summarize respondent profiles, current practices and proposed features. Elderly respondents strongly agreed that caregivers regularly checked their health vitals (M = 4.33). Caregivers reported feeling overwhelmed when monitoring multiple patients (M = 4.61) and considered manual monitoring time-consuming (M = 4.52), while their ability to monitor all patients' vitals received a neutral rating (M = 3.04). Most caregivers reported familiarity with work-related mobile applications, and many elderly respondents reported comfort wearing a smartwatch, although concern about all-day wear remained evident (M = 4.14). Caregivers strongly agreed with proposed system mechanisms (reported grand mean = 4.53) and key features (reported grand mean = 4.58); chatbot guidance obtained the highest mechanism rating (M = 4.64). The results indicate perceived demand for organized monitoring, caregiver notifications, AI assistance and reporting. They support a supervised pilot and further usability and reliability evaluation, but do not establish improved clinical outcomes, sensor accuracy or emergency response time. ElderCareAI is intended to complement existing caregiving practices and professional judgment.

Keywords: elderly monitoring, caregiver support, wearable technology, Galaxy Watch6, artificial intelligence, ElderCareAI